Directions: Answer the following questions and circle the corresponding option.

$$8, -4, 2, -1, \ldots$$

1. The first four terms of the geometric sequence $g_n = g_0(r)^n$ are shown above. What is the value of r?

(a) $r = 2$ (b) $r = -2$ (c) $r = \frac{1}{2}$ (d) $r = -\frac{1}{2}$

2. Let a_n be an arithmetic sequence where $a_3 = 5$ and $a_9 = 14$. What is the value of a_{12}?

(a) 18 (b) 18.5 (c) 19 (d) 23

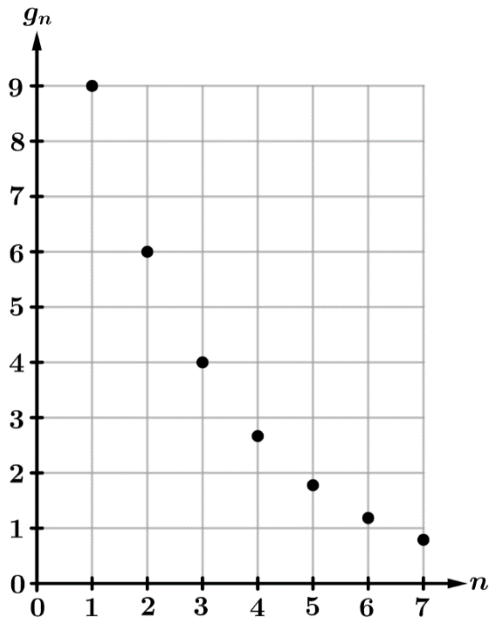
3. Let g_n be a geometric sequence where $g_4 = -5$ and $g_6 = -45$. Which of the following could be r?

I. $r = -3$ II. $r = 3$ III. $r = 9$

(a) I only (b) II only (c) III only (d) I and II

4. Let a_n and g_n be an arithmetic and geometric sequence with $a_1 = g_1 = -1$ and $a_4 = g_4 = 8$. What is the value of $a_5 - g_5$?

(a) -5 (b) 0 (c) 5 (d) 27



5. Values of the terms of a geometric sequence g_n are graphed in the figure above. Which of the following is an expression for the nth term of the geometric sequence?

(a) $g_n = 4\left(\frac{3}{2}\right)^{n-2}$

(b) $g_n = 6\left(\frac{2}{3}\right)^{n-2}$

(c) $g_n = 9\left(\frac{2}{3}\right)^{n}$

(d) $g_n = 9\left(\frac{3}{2}\right)^{n-1}$